

TRUSTLENS: Build-Ready Technical Research Report

TL;DR

- **Build TrustLens as an OpenAI-compatible middleware proxy (via LiteLLM) plus a small verification pipeline, not a fine-tuned model** — the drop-in `base_url` override is the only integration surface that genuinely works with "any AI" in 24 hours; wrap a fast grounded-factuality checker (LettuceDetect ModernBERT for span output, or MiniCheck for claim verdicts) around it.
- **Span-level "correct vs hallucinated" highlighting already exists** in LettuceDetect (open-source, MIT, token-level, RAGTruth-trained) and in the RAGTruth dataset itself — do NOT rebuild these; TrustLens's genuine white space is the *combination* nobody ships together: claim-level verdicts + supporting AND contradicting evidence + source-quality scoring + risk-adaptive depth + agent action-gating via an MCP server.
- **Use RAGTruth as your primary dataset** (span-level human hallucination annotations, MIT license) with HaluBench and LLM-AggreFact for eval; current SOTA fact-checkers top out around 77% balanced accuracy, so calibrate all claims — this is an assistive, not authoritative, verifier.

Key Findings

1. **The competitive table is mostly accurate but two ratings are understated.** "Claim-level verification" and "Evidence for every claim" are rated *Partial* for existing tools — correct, but RAGAS, TruLens, DeepEval, and DeepMind's SAFE already decompose responses into claims and check each. So TrustLens's edge on those two rows is *presentation* (span highlighting + supporting/contradicting evidence + provenance), not the raw capability. "Supporting + contradicting evidence" as CORE is genuinely defensible white space — almost no shipping tool deliberately mines *contradicting* evidence rather than only confirming.
2. **Span-level highlighting is a solved sub-problem.** LettuceDetect returns character offsets of hallucinated spans out of the box (`{start, end, confidence, text}`). Rebuilding this from scratch is a time sink; wrap it instead.
3. **The most defensible 24-hour demo** is the OpenAI-compatible proxy + a Streamlit/Gradio UI showing color-coded spans, plus one "wow" differentiator: an MCP server that blocks an agent tool call until its supporting claims verify.
4. **Cheap per-query verification is achievable:** encoder models (LettuceDetect, MiniCheck-RoBERTa, HHEM-2.1) run on CPU/T4 for near-zero marginal cost; web-search verification costs roughly one search per claim.

Details

SECTION 1 — Landscape / Competitive Analysis

Hallucination detection / grounding models (grounded factuality = "is this claim supported by THIS context"):

- **Vectara HHEM-2.1-Open** (`vectara/hallucination_evaluation_model`), HuggingFace): 110M-param classifier, outputs a 0-1 factual-consistency score. Runs on CPU, <600MB RAM at 32-bit, ~1.5s for a 2k-token input on a modern x86 CPU. Open weights (Apache-2.0). Whole-response/claim granularity, no span offsets, no evidence surfaced, no source scoring. Trained partly on the RAGTruth training set. Downloaded 3.5M+ times. Scores ~71.8% balanced accuracy on LLM-AggreFact.
- **Bespoke-MiniCheck-7B** (`bespokelabs/Bespoke-MiniCheck-7B`): Tops the public LLM-AggreFact leaderboard at **77.4% balanced accuracy, surpassing models many times larger** (`Bespokelabs`) (per Bespoke Labs docs), with **~200ms end-to-end latency on a single modern GPU (<100ms with optional optimisations)** and **over 500 documents/minute**. (`PromptLayer`) Binary supported/unsupported per sentence. **License: free for non-commercial use only; commercial licensing requires contacting Bespoke Labs (`company@bespokelabs.ai`)**. Fine-tuned from internlm2_5-7b-chat (`Hugging Face`) on 35K examples. (`Hugging Face`)
- **MiniCheck smaller variants** (`lytang/MiniCheck-Flan-T5-Large`), MiniCheck-RoBERTa-Large): Flan-T5 hits 74.7% balanced accuracy — matching Claude-3 Opus (74.1%) and approaching GPT-4 (75.3%) — at ~400x lower inference cost (\$0.24 vs \$107 on the 13K test set). (`alphaXiv`) RoBERTa-Large (355M) ~73% and CPU-friendly.
- **Patronus Lynx** (`PatronusAI/Llama-3-Patronus-Lynx-8B-Instruct`), 70B variant, and a GGUF-quantized 8B for Ollama): LLM-based RAG hallucination judge, produces PASS/FAIL + chain-of-thought reasoning. (`arXiv`) Open; integrated natively into NVIDIA NeMo Guardrails. (`NVIDIA`) Evaluated on HaluBench; (`arXiv`) the 8B "Lynx 2.0" reportedly detects 8 hallucination subtypes (coreference, calculation, CoT, etc.). (`Patronus`)
- **AlignScore** (`yzha/AlignScore`): 355M, ~71-72% on LLM-AggreFact. Older baseline.
- **LettuceDetect** (`KRLabsOrg/lettucedect-large-modernbert-en-v1`) and (`-base-modernbert-en-v1`): **the key find — MIT license, ModernBERT token-classification, returns span-level character offsets of hallucinated text**. (`lettucedetect-large-v1`) attains **79.22% example-level F1 on the RAGTruth test set, outperforming prompt-based GPT-4 (63.4%), encoder-based Luna (65.4%), and fine-tuned Llama-2-13B (78.7%)** (`Hugging Face`) (LettuceDetect paper, arXiv 2502.17125). pip-installable (`pip install lettucedetect`), 4K context (ModernBERT) / 8K (EuroBERT multilingual variants: base/210M, large/610M). (`GitHub`) This is exactly the "which parts are hallucinated" feature the user described.
- **Cleanlab TLM**: closed API, scores trustworthiness 0-1 for ANY LLM output, model-agnostic, (`Cleanlab`) returns explanations (e.g. "Found alternate plausible response:

December"). ([GitHub](#)) Good reference for the "confidence score" UX and the model-agnostic positioning, but not open. Does not natively stream both response and score. ([LlamaIndex](#))

- **Amazon Bedrock Guardrails** (Automated Reasoning checks, contextual grounding checks) and **Azure AI Content Safety Groundedness Detection** (with a correction capability): closed cloud services offering grounding + correction — competitors on the "guardrails" row.

Observability/tracing (LangSmith, Langfuse, Arize Phoenix, W&B Weave, Braintrust, HoneyHive): strong category; TrustLens correctly says "not a focus."

Guardrail frameworks: NVIDIA NeMo Guardrails (integrates Patronus Lynx natively via vLLM/Ollama), ([NVIDIA](#)) Guardrails AI, Llama Guard, Invariant Labs. TrustLens can sit alongside these rather than compete.

Evaluation frameworks: RAGAS (faithfulness metric decomposes answer into claims, checks each against context; integrates HHEM-2.1-Open), TruLens (RAG Triad: context relevance, groundedness/faithfulness, answer relevance, with LLM-as-judge + supporting reasoning), ([arxiv](#)) DeepEval (FaithfulnessMetric emphasizes contradiction detection), ([arxiv](#)) promptfoo, Giskard. These already do claim-level checking — so the table's "Partial" is generous to TrustLens; differentiation must be UX + evidence + provenance + gating. Practical threshold from the field: faithfulness <0.85 is a regression, <0.75 means the bot is reliably fabricating. ([DEV Community](#))

Automated fact-checking research systems: FActScore (atomic-fact decomposition + per-fact support), **SAFE / Search-Augmented Factuality Evaluator** (Google DeepMind, [google-deepmind/long-form-factuality](#)): splits response into atomic facts, decontextualizes each, checks via Google Search; **agrees with 72% of ~16k human annotations, wins 76% of 100 sampled disagreement cases, and is >20× cheaper than human annotators** — Wei et al., arXiv 2403.18802), FacTool, RARR, Chain-of-Verification (CoVe), VeriScore, OpenFactCheck, Loki, Factcheck-GPT. These are the intellectual backbone of TrustLens's open-web mode.

Genuine white space for TrustLens: (1) deliberately surfacing *contradicting* evidence, not just confirming; (2) source-quality/provenance scoring bundled into the verdict; (3) risk-adaptive verification depth; (4) agent action-gating (blocking a tool call until claims verify) — genuinely "emerging" and the strongest differentiator. Span-level correct-vs-hallucinated highlighting is NOT white space (LettuceDetect/RAGTruth own it) — reuse, don't rebuild.

SECTION 2 — Technical Methods

(a) Claim decomposition. Cheapest reliable path in hours = an LLM prompt implementing FActScore/SAFE-style atomic-fact extraction + decontextualization. **Claimify** (Microsoft Research, arXiv 2502.10855) is the strongest method — 3 stages (Selection → Disambiguation → Decomposition), ([MarkTechPost](#)) 99% entailment rate, 87.6% coverage,

(MarkTechPost) and uniquely refuses to extract claims when the source sentence has multiple plausible interpretations. An unofficial implementation exists (github.com/deshwalmaresh/claimify). Decontextualization (resolving pronouns/references so each claim is self-contained) is essential — otherwise retrieval fails. For 24h: sentence-split + LLM decontextualization prompt is adequate; full Claimify if time permits.

(b) Evidence retrieval. Search APIs — **Tavily** is the best hackathon default: the free Researcher tier grants **1,000 API credits every month, no credit card required** (basic search = 1 credit, advanced = 2; (Tavily Docs) pay-as-you-go \$0.008/credit), (Tavily Docs) and returns cleaned content (it's the LangChain default). Serper (2,500 trial queries, cheap raw Google SERP, \$50/mo for 50K), (BuildMVPFast) Exa (1,000/month, semantic/neural search + findSimilar), Brave (free tier removed early 2026, now ~\$5/month credit ≈ 1,000 queries). (CodeNote) Note: Bing Search API was deprecated 11 Aug 2025; Google Custom Search JSON API is closed to new signups and discontinues 1 Jan 2027. (Parallel) Local retrieval: embeddings (BAAI/bge-small-en-v1.5), (sentence-transformers/all-MiniLM-L6-v2) (fast/free/CPU), GTE, E5, Nomic; vector stores FAISS (fastest to stand up), Chroma, Qdrant, LanceDB. Hybrid BM25+dense improves recall. Offer two modes: RAG-over-provided-context (grounding check) vs open-web verification. Caution: retrieved web pages are an indirect prompt-injection vector. (WebCraft)

(c) Claim-evidence entailment / verdict.

Model	HF ID	Balanced Acc (LLM- AggreFact)	Notes
Bespoke-MiniCheck-7B	bespokelabs/Bespoke-MiniCheck-7B	77.4%	non-commercial license, GPU, ~200ms
MiniCheck-Flan-T5-Large	lytang/MiniCheck-Flan-T5-Large	74.7%	GPT-4 level, ~400× cheaper, CPU-ok
MiniCheck-RoBERTa-L	(Liyan06/MiniCheck repo)	~73%	355M, CPU-friendly
HHEM-2.1-Open	vectara/hallucination_evaluation_model	~67-72%	110M, CPU, 1.5s/2k tokens, Apache-2.0
AlignScore-large	yzha/AlignScore	~71-72%	355M
Lynx-8B	PatronusAI/Llama-3-Patronus-Lynx-8B-Instruct	strong, gives reasoning	LLM judge, GGUF via Ollama
DeBERTa-v3 NLI	MoritzLaurer/mDeBERTa-v3-base-mnli-xnli , cross-encoder/nli-deberta-v3-base	—	zero-shot 3-way NLI, CPU

Recommendation: Default = **LettuceDetect large** (span output + MIT + CPU) OR **MiniCheck-Flan-T5-Large** (claim verdicts + permissive license). Fallback/free-hardware = HHEM-2.1-Open (CPU, tiny) or a DeBERTa-v3 NLI cross-encoder. Avoid Bespoke-MiniCheck-7B in the shipped demo unless you clear its non-commercial license and have a GPU.

(d) Supporting vs contradicting (3-way verdict). Use the **FEVER label scheme**: **SUPPORTS** / **REFUTES** / **NOT ENOUGH INFO**. NLI models natively output entailment/contradiction/neutral → map directly. To deliberately mine *contradicting* evidence (combat confirmation bias), generate counter-queries ("evidence that X is false", negation-rewrites), run stance detection on retrieved passages, and retrieve for both the claim and its negation. Confirmation-bias failure mode: if you only search the claim as

written, you retrieve confirming sources and label everything SUPPORTED — a real and common defect this feature exists to fix.

(e) Confidence / calibration. Signals — token logprobs/sequence probability; **semantic entropy** (Farquhar, Kossen, Kuhn, Gal, *Nature* 630:625–630, 2024 [\(Nature\)](#) — clusters semantically equivalent samples then computes entropy; detects confabulations without ground truth); [\(arxiv\)](#) **semantic entropy probes** (arXiv 2406.15927 — approximate semantic entropy from hidden states of a single generation, near-zero overhead); [\(arXiv\)](#) SelfCheckGPT (sampling consistency); P(True) self-eval; verbalized confidence (poorly calibrated — don't trust the model saying "95% sure"); conformal prediction for abstention. **Why LLMs are confidently wrong:** OpenAI's "Why Language Models Hallucinate" (Kalai, Nachum, Vempala, Zhang; arXiv 2509.04664, Sept 2025) argues training/eval procedures reward guessing over abstention, binary scoreboards penalize "I don't know," and the generative error rate is provably $\geq 2\times$ the classification error rate. [\(Substack\)](#) This is the causal grounding for TrustLens's value proposition. Express confidence in the UI as discrete bands (High/Medium/Low/Unverifiable), never false-precision percentages.

(f) Source quality / provenance. Media Bias/Fact Check (11,000+ sources, 0–5 factuality score / Very-High→Very-Low, a paid Data API plus scrapable ratings, volunteer methodology) is the most complete openly available set of source-veracity labels; **NewsGuard** (0–100 trust score across 9 criteria) [\(arxiv\)](#) is higher quality but **not free/licensed** — do not scrape it. [\(arxiv\)](#) Also Ad Fontes Media, the Wikipedia Reliable Sources perennial list (scrapable), domain authority, journal/venue signals, recency/staleness, primary-vs-secondary detection, and C2PA content-provenance standards. For a hackathon, build a defensible heuristic rubric: a static allow/deny list (Wikipedia perennial + a small MBFC-derived table) + domain-type weighting (gov/edu/peer-reviewed > mainstream news > blog/forum > known-unreliable) + recency + corroboration count. Research datasets like NELA-GT bundle MBFC/NewsGuard-derived labels for reference. [\(arxiv\)](#)

(g) Risk-adaptive verification. Classify each query/claim by domain risk (medical/legal/financial/safety-critical vs trivia) with a keyword+LLM router, then cascade: cheap encoder check first → escalate to web search + LLM judge + human only for high-risk or low-confidence claims. This is cost-aware routing / selective prediction. Set verification depth and latency budgets per tier.

(h) Human escalation & audit trail. Review queue for low-confidence/high-risk verdicts. The audit record should contain: original prompt, model + version, response, extracted claims, retrieved evidence (URLs + snippets + timestamps), per-claim verdicts + scores, verifier model versions, final decision, and human reviewer action. Compliance drivers: **EU AI Act Article 50 transparency obligations apply from 2 August 2026, with fines up to €15M or 3% of worldwide annual turnover (Art. 99(4)), whichever is higher** (final Commission guidelines adopted 20 July 2026); Article 50(2) machine-readable marking of AI content has a transition to 2 December 2026. [\(Disclosed\)](#) Most high-risk obligations were delayed to Dec 2027/Aug 2028 by the June 2026 Digital Omnibus, but Article 50

transparency landed on schedule. (Digital Applied Team) Also NIST AI RMF and ISO/IEC 42001.

(i) **Decision/action gating for agents (the major opportunity)**. Tool-calling agents get gated today via OpenAI Agents SDK approval policies, (OpenAI) Anthropic tool-use approval, **MCP elicitation / human-in-the-loop approval gates**, **LangGraph interrupts** (surface an approval round mid-execution), (LangChain) AutoGen/CrewAI guardrails, and Invariant Labs agent security. **TrustLens can ship as an MCP server**: it exposes a (verify_before_action) tool; the agent host requires the verification verdict (or approval) before executing a sensitive (tools/call). Concretely: intercept the proposed action + its justifying claims → verify claims → if any REFUTED/unsupported above the risk threshold, block/return failure or route to a human. MCP is model-agnostic (works with OpenAI, Gemini, Ollama/vLLM, Anthropic), (Kunal Chowdhury) and FastMCP makes the server fast to build. (LangChain)

SECTION 3 — Architecture Decision

Recommendation: API middleware/proxy (primary) + agentic verification pipeline (internal) + MCP server (differentiator). NOT a fine-tuned model.

Rationale: The user wants to "integrate with any AI without issues." The only integration surface that delivers this in 24h is an **OpenAI-compatible proxy**. **LiteLLM** exposes a self-hosted OpenAI-compatible gateway — any client that uses the OpenAI SDK works by changing (base_url) to your proxy, with no code changes, across 100+ providers (LiteLLM) (OpenAI, Anthropic, Gemini, Bedrock, Ollama, vLLM, HuggingFace, etc.). TrustLens performs **post-hoc verification** after the model response completes (streaming + inline verification is hard — do post-hoc).

Integration surfaces assessed for 24h feasibility:

- OpenAI-compatible proxy (LiteLLM) — **YES, do this**, most defensible.
- SDK wrapper — easy but less "drop-in."
- Browser extension (DOM injection into ChatGPT/Claude/Gemini web UIs) — flashy but brittle and time-consuming; skip unless a teammate owns it.
- MCP server — **YES for the agent-gating differentiator** (FastMCP).
- OpenRouter as the upstream provider — fine.

Affordability. Encoder verifiers (LettuceDetect/MiniCheck/HHEM) run on CPU or a Colab T4 at ~zero marginal cost; cache verdicts by (claim, evidence) hash; cascade cheap-first; batch claims; use free-tier search (Tavily 1,000/mo). Rough per-query cost: context-grounded mode (no web search) ≈ near-zero (local model only); open-web mode ≈ 1-2 Tavily credits (~\$0.008-0.016 at PAYG) + a small LLM call per claim → a few cents per multi-claim answer. Free LLM tiers (Groq, Gemini, Together) keep the hackathon bill at \$0.

SECTION 4 — Span-Level Output Format

Map atomic claims back to character offsets via offset mapping; when claims are decontextualized (no longer verbatim), use fuzzy alignment (token overlap / embedding similarity / rapidfuzz) to locate the source span. **RAGTruth** annotates hallucination spans with char `(start)`/`(end)`, `(text)`, and a `(label_type)` `(GitHub)` (Evident/Subtle × Conflict/Baseless-Info). LettuceDetect returns `{start, end, confidence, text}` directly. `(Hugging Face)`

Recommended label taxonomy: **Supported** / **Contradicted** / **Unsupported-no-evidence** / **Opinion-or-subjective** / **Not-checkworthy**.

Published UX risks to design around: **over-trust of highlighted output, alert fatigue, and the "unhighlighted = correct" fallacy** (users assume anything not flagged is verified). Mitigate by surfacing per-claim confidence, labeling coverage explicitly ("3 of 5 sentences were checkable"), and never implying unhighlighted text is verified-correct.

Recommended JSON schema for a verification result:

```
json
{
  "response_id": "uuid",
  "model": "provider/model",
  "overall_trust": {"band": "Medium", "score": 0.62},
  "coverage": {"checkable_claims": 5, "checked": 5},
  "claims": [
    {
      "claim_id": 1,
      "text": "decontextualized claim",
      "source_span": {"start": 0, "end": 42, "text": "verbatim span in original"},
      "verdict": "SUPPORTED | REFUTED | NOT_ENOUGH_INFO | OPINION | NOT_CHECKWO",
      "confidence": 0.87,
      "evidence": [
        {"stance": "supporting|contradicting", "url": "...", "snippet": "...",
          "source_quality": {"band": "high", "score": 0.8}, "retrieved_at": "ISO"}
      ]
    }
  ],
  "audit": {"verifier_models": ["KRLabsOrg/lettucedect-large-modernbert-en-v1"]}
}
```

SECTION 5 — Datasets (HIGH PRIORITY)

PRIMARY: RAGTruth. Original at `(github.com/ParticleMedia/RAGTruth)`
`(dataset/response.jsonl) + (dataset/source_info.jsonl)`, join on `(source_id)`, **MIT license**.

2,965 prompts → 17,790 responses from six LLMs (GPT-3.5-Turbo-0613, GPT-4-0613, Llama-2-7B/13B/70B-Chat, Mistral-7B-Instruct), ([arxiv](#)) ([github](#)) ~15,090 train / 2,700 test, ([arxiv](#)) with 14,289 human-annotated hallucination spans ([github](#)) (char [start](#)/[end](#)/[text](#)/[label_type](#)), across tasks QA / Data2txt / Summary (Niu et al., ACL 2024). [response.jsonl](#) fields: ([id](#)), ([source_id](#)), ([model](#)), ([temperature](#)), ([labels](#)) (list of span dicts), ([split](#)), ([quality](#)), ([response](#)). [github](#) [source_info.jsonl](#) fields: ([source_id](#)), ([task_type](#)), ([source](#)), ([source_info](#)), ([prompt](#)). [github](#) No native [load_dataset\(\)](#) — download the two JSONL files. HF derivative with token logits: ([jakobsnel/RAGTruth_Xtended](#)) ([Hugging Face](#)) ([load_dataset\("jakobsnel/RAGTruth_Xtended"\)](#)). **Reason it's primary:** the only large public corpus with human span annotations — exactly TrustLens's core feature.

SECONDARY / EVAL:

- ([PatronusAI/HaluBench](#)) — ([load_dataset\("PatronusAI/HaluBench"\)](#)), split ([test](#)), **14,900 examples**, ([huggingface](#)) fields ([id](#) / [passage](#) / [question](#) / [answer](#) / [label](#) ([PASS](#)|[FAIL](#)) / [source_ds](#)). **License cc-by-nc-2.0 (non-commercial)**. Caveat: repackages RAGTruth/HaluEval/PubmedQA/CovidQA/DROP/FinanceBench ([Hugging Face](#)) → contamination risk if you also train on those. The HaluEval-only subset is ([flowai.com/HaluEval](#)) ([Hugging Face](#)) (10,000 rows).
- ([lytang/LLM-AggreFact](#)) — ([load_dataset\("lytang/LLM-AggreFact", token=True\)](#)) (**GATED** — must log in and accept conditions), ([Hugging Face](#)) dev 30,420 / test 29,320, fields ([dataset](#) / [doc](#) / [claim](#) / [label](#) (1|0) / [contamination_idenfier](#)). Best for benchmarking your verifier via balanced accuracy per sub-dataset.
- ([pminervini/HaluEval](#)) — configs ([qa](#) / [dialogue](#) / [summarization](#) / [general](#)) (+[_samples](#)), ~64.5k rows, ([Hugging Face](#)) one ([data](#)) split each; RUCAIBox origin, MIT. Fields per task (QA: ([knowledge](#) / [question](#) / [right_answer](#) / [hallucinated_answer](#))).
- **FEVER** — ([load_dataset\("fever", "v1.0"\)](#)), labels ([SUPPORTS](#) / [REFUTES](#) / [NOT ENOUGH INFO](#)), ~427k rows (train 311k), ([Hugging Face](#)) CC BY-SA 3.0. Fields: ([id](#), [label](#), [claim](#), [evidence_wiki_url](#), [evidence_sentence_id](#)), ([TensorFlow](#)) etc. For training/using 3-way entailment. Caveat: ([NonMatchingChecksumError](#)) on some ([datasets](#)) versions ([GitHub](#)) — use an updated ([datasets](#)) or ([verification_mode="no_checks"](#)). Variants: Climate-FEVER, FEVEROUS, SciFact, PubHealth.
- **FaithBench** ([github.com/vectara/FaithBench](#)), JSON batches 1-16 except 13, ([GitHub](#)) no HF loader) — summarization, 4 classes Consistent/Benign/Questionable/Unwanted with span offsets; deliberately "challenging" cases ([arXiv](#)) so not representative of base rates; verify repo LICENSE before redistribution.
- **TruthfulQA** — ([truthfulqa/truthful_qa](#)), Apache-2.0, ([Hugging Face](#)) 817 questions across 38 categories ([Hugging Face](#)) — for a closed-book truthfulness demo.
- **Multilingual / Indian:** ([utahnlp/x-fact](#)) (MIT, 31,189 claims, 25 languages including Hindi, Marathi, Tamil, Bengali, Gujarati, Punjabi; 7-way labels true→false +

unverifiable). Indian-specific FactDrill (13 Indian languages) and IFND exist academically but no clean HF loader was confirmed.

- **Span-derivative for training** (English, summarization): (`llm-semantic-router/halueval-llm-spans`) (`Hugging Face`) (RAGTruth-format spans, Apache-2.0) — but LLM-annotated → label noise.
- **NLI training/inference**: MNLI, ANLI, VitaminC.

SECTION 6 — 24-Hour Build Feasibility

Buildable in 24h: LiteLLM proxy (pip, OpenAI-compatible) + LettuceDetect (`pip install lettucedetect`) for span output + Tavily search (free 1,000/mo) + a Streamlit or Gradio UI + FastMCP for the agent-gating demo. Free hardware: Colab T4; CPU is fine for encoders (HHEM/LettuceDetect/MiniCheck-RoBERTa/DeBERTa). Free LLM tiers: Groq, Gemini, Together.

Time sinks to avoid: browser-extension DOM injection; streaming inline verification; full Claimify; any fine-tuning; scraping licensed source-quality data; the FEVER checksum bug.

Metrics to report in the demo: span-level F1 (vs RAGTruth), claim-level precision/recall, balanced accuracy (vs LLM-AggreFact/HaluBench), AUROC for the trust score, latency p50/p95, and cost per query. **Baselines to beat**: HHEM-2.1-Open, an LLM-as-judge (GPT-4o-mini) faithfulness prompt, and RAGAS faithfulness.

SECTION 7 — Risks and Honest Limitations

- **Ceiling of current fact-checking**: SOTA grounded factuality is ~77% balanced accuracy (Bespoke-MiniCheck 77.4%, MiniCheck-FT5 74.7%, GPT-4 75.6%, (`arxiv`) HHEM-2.1 ~71.8%); LettuceDetect ~79.2% example-level F1 on RAGTruth; SAFE ~72% agreement with humans. So roughly 1 in 4–5 verdicts can be wrong. Present TrustLens as assistive, not authoritative.
- **Verifying the verifier**: the checker itself errs; report its benchmarked accuracy honestly in the UI.
- **Retrieval failure → false "unsupported"**: if search misses the evidence, a true claim gets flagged. Always distinguish REFUTED (found contradicting evidence) from NOT_ENOUGH_INFO (found nothing).
- **The web contains misinformation, AI-generated slop, and circular citations** — source-quality scoring mitigates but does not solve this.
- **Unverifiable-in-principle, subjective/opinion, and time-sensitive claims** — must be labeled Opinion / Not-checkworthy, never forced into true/false.
- **Confirmation bias in retrieval** — mine contradicting evidence deliberately (Section 2d).
- **Contamination** across HaluBench / LLM-AggreFact / RAGTruth — avoid train/test leakage.

Recommendations

Stage 1 (hours 0-6): Core pipeline. Stand up the LiteLLM OpenAI-compatible proxy (`(base_url)` override). Wire LettuceDetect-large for span-level grounding against provided context. Ship a Gradio/Streamlit UI with color-coded spans (green = supported, red = contradicted, grey = unsupported). This alone is a working demo of the user's exact ask.

Stage 2 (hours 6-14): Evidence + verdict + provenance. Add Tavily open-web mode; add claim decomposition (LLM decontextualization prompt); add 3-way FEVER-scheme verdicts via a DeBERTa NLI cross-encoder or MiniCheck; add counter-query generation for contradicting evidence; add the heuristic source-quality rubric. Emit the JSON schema output + audit record.

Stage 3 (hours 14-20): Differentiators. Ship the MCP server with a `(verify_before_action)` gate (FastMCP + a LangGraph interrupt demo). Add risk-adaptive routing (keyword/LLM risk classifier → cascade depth). Add confidence bands.

Stage 4 (hours 20-24): Eval + polish. Run span-level F1 on the RAGTruth test set + balanced accuracy on a HaluBench sample; produce a metrics slide (F1, balanced accuracy, latency p50/p95, cost/query) beating HHEM / LLM-judge baselines. Record the demo video.

Thresholds that change the plan: if no GPU → drop Bespoke-MiniCheck-7B, use HHEM-2.1 / DeBERTa on CPU. If Tavily credits run out → switch to the Serper trial or a local corpus + FAISS. If span-alignment for decontextualized claims proves flaky → fall back to sentence-level highlighting. If time is short → cut the browser extension and open-web mode, keep proxy + LettuceDetect + UI.

Caveats

- Benchmark numbers are drawn from each model's own papers/leaderboards (LLM-AggreFact, RAGTruth) and may be optimistic versus your data distribution; re-measure on your own eval set.
- Free-tier API limits and Brave/Bing/Google availability shift frequently (Bing deprecated Aug 2025, Google CSE closing Jan 2027, `(Parallel)` Brave perpetual free tier retired early 2026, `(CodeNote)` Tavily acquired by Nebius Feb 2026); `(Dennisworks)` re-verify at build time.
- Bespoke-MiniCheck-7B, HaluBench, and HaluEval-derived sets carry non-commercial license restrictions — check before any commercial use or public release.
- The EU AI Act Article 50 timeline reflects the June/July 2026 Digital Omnibus amendments and guidelines; confirm current status if compliance is part of your pitch.